Market History, Market Cycle Theory and Market Cycle Indicators

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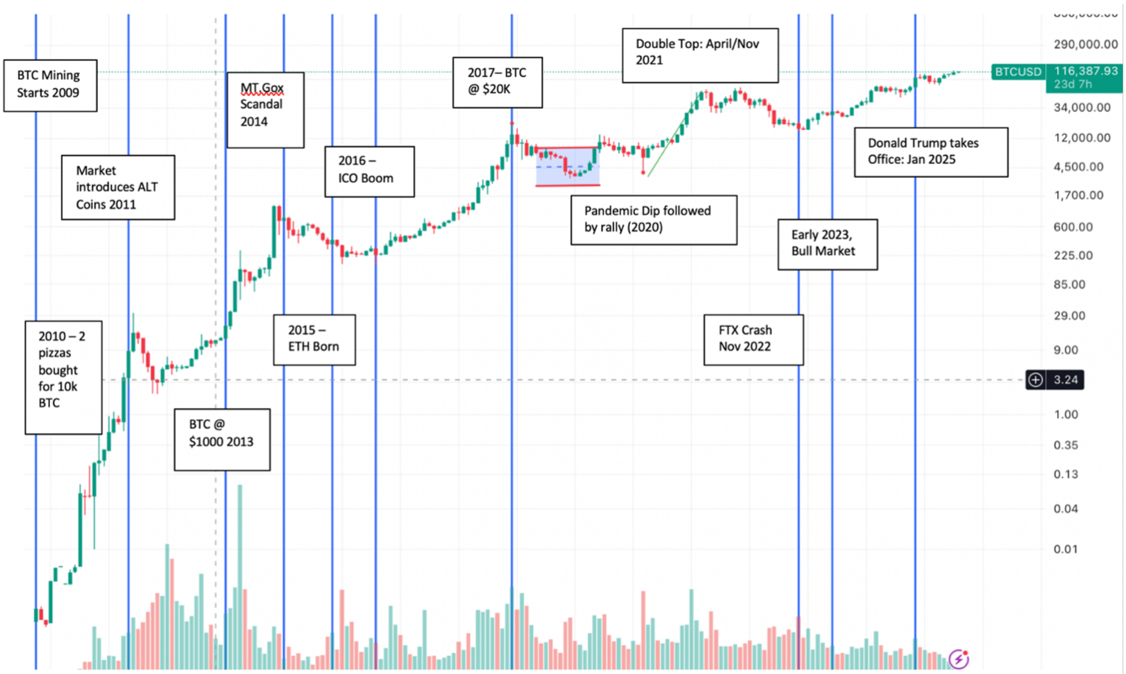
Market History



2009: Genesis Block & Early Mining

- **Event:** January 3, 2009 – Satoshi Nakamoto mined the first Bitcoin block, known as the Genesis Block.
- **Context:** Bitcoin's inception marked the birth of decentralized digital currency, introducing a new paradigm in financial systems.

2010: First Real-World Transaction

- **Event:** May 22, 2010 – Laszlo Hanyecz purchased two pizzas for 10,000 BTC.
- **BTC Price:** Approximately $0.0025 per BTC.
- **Context:** This transaction is celebrated as "Bitcoin Pizza Day," symbolizing the first real-world use of Bitcoin.

2011: Emergence of Altcoins

- **Event:** 2011 – Introduction of alternative cryptocurrencies (altcoins).
- **BTC Price:** Ranged from $1 to $30.
- **Context:** The rise of altcoins like Litecoin diversified the cryptocurrency market, offering alternatives to Bitcoin.

2013: Bitcoin Hits $1,000

- **Event:** Late 2013 – Bitcoin's price surged to over $1,000.

- **BTC Price:** Reached approximately $1,100 in November 2013.
- **Context:** Factors contributing to the surge included increased media attention, adoption by online retailers, and growing interest from investors.

2014: Mt. Gox Exchange Collapse

- **Event:** February 2014 – Mt. Gox, the largest Bitcoin exchange at the time, filed for bankruptcy after losing approximately 850,000 BTC.
- **BTC Price:** Dropped from around $1,000 to below $400.
- **Context:** The collapse highlighted security vulnerabilities in cryptocurrency exchanges, leading to increased regulatory scrutiny.

2015: Ethereum Launch

- **Event:** July 2015 – Ethereum, a new blockchain platform enabling smart contracts, was launched.
- **BTC Price:** Approximately $250.
- **Context:** Ethereum's introduction expanded the use cases of blockchain technology beyond digital currency.

2016: ICO Boom

- **Event:** 2016 – Initial Coin Offerings (ICOs) became a popular method for blockchain projects to raise funds.
- **BTC Price:** Ranged from $400 to $900.
- **Context:** The ICO boom led to a surge in new cryptocurrency projects but also attracted regulatory attention due to concerns over scams and investor protection.

2017: Bitcoin Reaches $20,000

- **Event:** December 2017 – Bitcoin's price peaked at nearly $20,000.
- **BTC Price:** Reached approximately $20,000.
- **Context:** The surge was driven by retail investor enthusiasm, media coverage, and the launch of Bitcoin futures trading.

2020: Pandemic-Induced Rally

- **Event:** March 2020 – Bitcoin's price dipped below $4,000 due to global market panic.
- **BTC Price:** Dropped to around $3,800.
- **Context:** The pandemic-induced economic uncertainty led to a flight to digital assets, with Bitcoin rebounding and reaching new highs.

2021: Double Top Formation

- **Event:** April and November 2021 – Bitcoin's price reached new all-time highs, forming a double top pattern.
- **BTC Price:** Peaked at approximately $64,000 in April and $69,000 in November.
- **Context:** The double top pattern indicated potential market exhaustion, leading to a subsequent correction.

2022: FTX Exchange Collapse

- **Event:** November 2022 – FTX, a major cryptocurrency exchange, filed for bankruptcy amid allegations of fraud.
- **BTC Price:** Dropped from around $20,000 to below $16,000.
- **Context:** The collapse of FTX shook investor confidence and highlighted the need for better regulation in the cryptocurrency industry.

2025: Market Recovery

- **Event:** August 2025 – Bitcoin's price rebounded to approximately $112,525.
- **BTC Price:** Around $112,525.
- **Context:** The recovery was fueled by renewed institutional interest, regulatory clarity, and broader adoption of blockchain technologies.

Market Cycle Theory

BTC itself trades and trends in a cyclical nature. This is similar to typical markets with their boom-and-bust cycles. These boom and bust cycles in traditional markets are typically correlated to monetary policy. Similarly, for BTC, the boom and bust cycles are highly correlated with the halving dates of BTC miner rewards. Left and right translation of market peaks for crypto markets is determined by monetary policy (will be examined further below).
In general, for BTC the cycle tends to be highly correlated with:

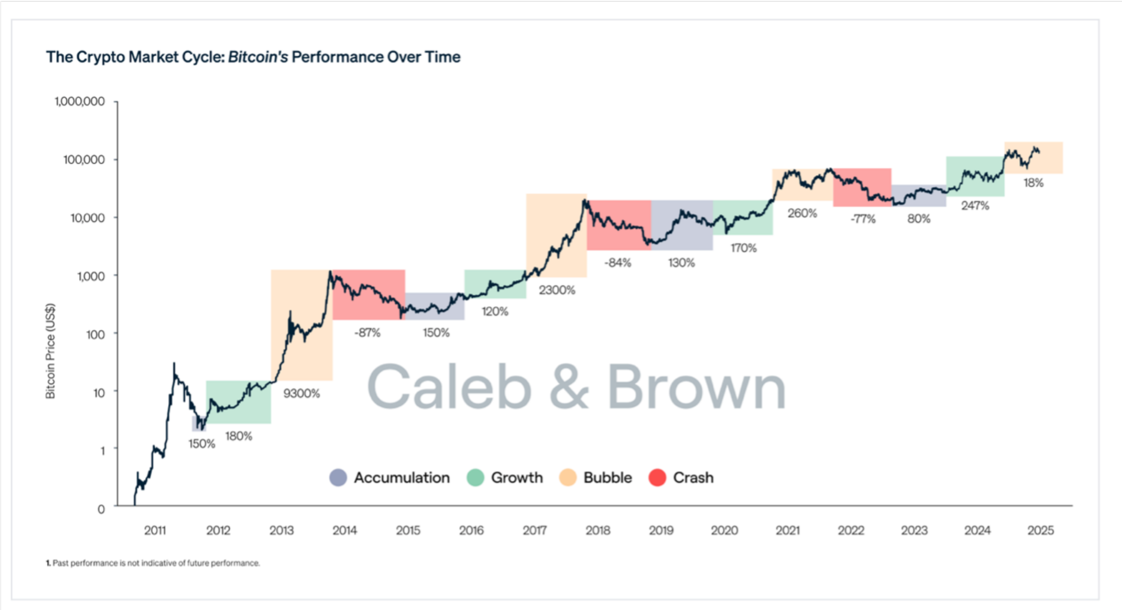
- Halving events
- Overall International Monetary Policy – Dictates left or right translated cycle
- Overall Market Fear and Greed

Different Market Stages:

1. Accumulation: Smart investors buy quietly after a prolonged downtrend; prices are low and sentiment is negative.

2. Markup / Early Bull: Price starts rising, early gains attract attention, volume increases.
3. Mania / FOMO: Rapid price surge; media hype and retail frenzy push BTC to euphoric highs.
   (a) Mark-up / Mania → Growth
4. Distribution: Early investors take profits; volatility rises, market peaks.
5. Decline / Bear: Prices fall, panic selling spreads, sentiment turns negative, setting up the next accumulation.
   (a) Distribution / Decline à Crash

Visualisation of the different Market Cycles: From the below images we

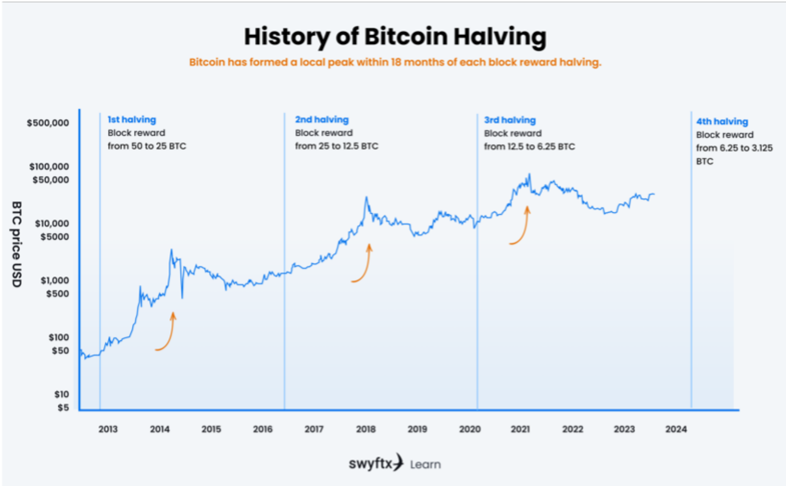


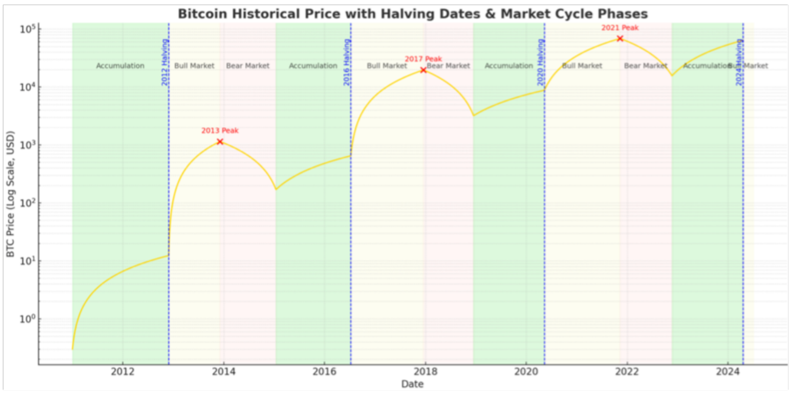
can also see the trends associated with the BTC Halving and the peaks of BTC. BTC has formed a local peak within 18-months of each block reward halving. This naturally ties into the different market cycle stages.

The Different Stages of Market Cycle Explained:

Stage 1 – Accumulation

- When? Usually occurs after a deep bear market when prices are consolidating at low levels.
- Investor psychology: Disbelief, apathy. Most retail investors are disinterested or have exited the market.
- On-chain signs:





- Long-term holders increase their BTC positions.
- Exchange reserves trend down as coins move to cold storage.
- Low trading volumes and volatility.

- Implication: Smart money and institutions accumulate here, anticipating future catalysts.

Stage 2 – Uptrend / Bull Market / Growth

- When? Often starts quietly, then accelerates after a major bullish catalyst (historically, the halving event is a key one).
- Investor psychology: Optimism → Euphoria. The "fear of missing out" (FOMO) drives inflows from retail traders.
- On-chain signs:
  - Rising active addresses and network usage.
  - Increased new wallet creation.
  - Higher exchange inflows/outflows as traders reposition.
- Market structure: Price makes higher highs and higher lows. Pullbacks are bought quickly.

- Implication: Potential for outsized gains but increasing risk as valuations rise.

Stage 3 – Distribution / Bubble

- When? Price has made substantial gains; market sentiment is euphoric.
- Investor psychology: Extreme greed. "This time is different" thinking dominates.
- On-chain signs:
  - Spikes in exchange inflows (holders taking profits).
  - Declining NUPL (Net Unrealized Profit/Loss) after a peak.
  - Whale wallets offloading to retail buyers.
- Market structure: Choppy sideways action near all-time highs, false breakouts, volatility spikes.
- Implication: Smart money begins to sell to late entrants; risk of a trend reversal grows.

Stage 4 – Downtrend / Bear Market

- When? After the top is in, price begins a sustained decline.
- Investor psychology: Complacency → Anxiety → Capitulation → Depression.
- On-chain signs:
  - Exchange reserves climb as people sell.
  - Realized losses increase.
  - Miner capitulation may occur if price < mining cost.
- Market structure: Lower highs and lower lows, occasional sharp relief rallies.
- Implication: Eventually sets up the next accumulation phase.

Why are these cycles repeating?

- In part due to supply shock associated with halving. Every 210,000 blocks (4 years), the block reward halves → reduces the rate at which new BTC enters circulation.
- This halves the rate of new BTC acting as a natural supply shock.
  - If sentiment / backdrop is available for spiked demand in combination with supply shock it causes price to increase.
- **12–18 months after each halving**, BTC has historically reached a cycle peak.

- This delay exists because:
  1.The supply shock is**gradual** – fewer coins trickle into the market daily.
  2.Demand tends to rise as bullish sentiment builds and mainstream awareness grows.
  - After the euphoric peak, speculative excess unwinds, triggering the bear phase.
  - Halving doesn't cause instant price spike, catalyst that tilts long-term supply demand balance and increases media attention. Human psychology magnifies the move.

The Mechanics of the Supply Shock:

- Pre-halving: Miners receive 6.25 BTC per block (~900 BTC/day).
- Post-halving: Rewards drop to 3.125 BTC (~450 BTC/day).
- That's ~450 BTC/day *less* being sold into the market.
- At $60,000/BTC, that's a ~$27M/day reduction in potential sell pressure.

Even though 19M+ BTC already exist, most are illiquid:

- Lost coins (~3-4M BTC)
- Long-term holders (HODLers) who rarely sell
- Corporate treasuries, ETFs, exchanges cold storage.
  This means only a small fraction of BTC trades actively — so removing even a modest flow of new coins tightens the available supply.

Psychological Aspect of Halving:

- Supply shock → pushes price up slightly → attracts media attention → increases retail & institutional demand → price rises more → demand compounds.
- This reflexive loop is why halvings often lead to *exponential* rather than *linear* price moves

Additional Factors Beyond Supply

1. Narrative Power – Halving is predictable & widely anticipated → traders front-run it → self-fulfilling prophecy.
2. Miner Economics – Post-halving, inefficient miners may shut down → reducing sell pressure further.
3. Market Sentiment – The halving often marks the *end of the bear* and start of the next bull narrative.
4. Macro Timing – If liquidity cycles line up with the halving, the effect can be amplified.

Left Translated Vs. Right Translated Market Cycles:

Average peak is 18 months after halving. Therefore, Left Translated is earlier than 18 months, right translated after 18 months (dictated by macro-climate) **Left-Translated Cycle**

- Peak occurs earlier in the cycle's timeframe.
- This means the uptrend is short-lived, followed by a longer downtrend/consolidation.
- Price tops before the mid-point of the cycle.

Implication for BTC:

- If a halving cycle is left-translated, it means:
  - The peak might happen sooner after the halving (e.g., within 6–12 months instead of 12–18).
  - The bear market and price drawdown phase will dominate most of the cycle.
  - Could indicate weaker demand or macroeconomic headwinds overpowering the halving-driven supply shock.

Right-Translated Cycle

- Peak occurs later in the cycle's timeframe.
- This means the uptrend is longer, followed by a shorter downtrend.
- Price tops after the mid-point of the cycle.
- Seen as bullish because the market spends more time rising.

Implication for BTC:

- If a halving cycle is right-translated:
  - The bull market may extend longer into the 4-year cycle.
  - The blow-off top might occur closer to the cycle's end (18–30 months post-halving).
  - Often associated with stronger adoption, liquidity inflows, and favorable macro conditions.

Alt Season and how it ties into Market Cycles:

Alt Season = Period when altcoins significantly outperform BTC.
Why it happens late in the cycle:

- Bitcoin rallies first after the halving → attracts new capital.
- As BTC's price growth slows, traders seek higher returns → rotate profits into alts.

- Momentum & hype push even low-quality projects up → speculative mania.
- Historically, alt season is one of the last signs before a market top.

Signals Alt Season may be starting:

1. BTC dominance declining after a sustained uptrend.
2. Large-cap alts (ETH, ADA, SOL) breaking out against BTC pairs.
3. Social media sentiment heavily skewed towards "cheap" alts.
4. Retail participation spikes (search trends, new exchange accounts, meme coins pumping).

Market Cycle Indicators

BTC Dominance as Market Cycle Indicator:

BTC Dominance is the portion of Crypto's market cap occupied by BTC.
Condensed Overview:

- BTC Leads market (safer asset), drives BTC dominance up
- As speculation/confidence grows, ALTCOINS become more attractive
- Altcoins Rally à Alt Asset price spike à BTC Dom goes down
- **Indicates over speculation/greed – redistribution phase à Indicates market cycle top**

"Inverse" to how Alt season indicates market top.

0.0.1 Detailed Overview as Part of Market Cycle Phases:

1. Early Bull Market (BTC-led phase)

- After a bear market, new liquidity first flows into BTC because it's the safest, most recognized, and most liquid crypto asset.
- Dominance usually **rises or stabilizes** during this phase, as BTC outperforms most alts.
- This phase typically corresponds to **Stage 1 & 2 accumulation and markup** in market cycle theory.

2. Mid Bull Market (Altcoin Rotation)

- Once BTC has made a big move up and its gains stabilize, investors start rotating into higher-risk assets (ETH, majors, mid-caps).
- BTC dominance begins to **fall** as altcoins catch up.

- Often corresponds to **late Stage 2 / early Stage 3** of the cycle.
- This is where ETH/BTC ratio starts to rise strongly.

3. Cycle Top (Alt Season / Distribution)

- Historically, **sharp drops in BTC dominance** (big altcoin rallies) have occurred near or just before market cycle tops.
- Example:
  - **2017:**BTC dominance fell from ∼85% → ∼37% by Jan 2018, just as the total market topped.
  - **2021:** BTC dominance peaked near ∼70% in Jan 2021, then collapsed to ∼40% as altcoins exploded into May 2021 — right before the market corrected.
- Why? Retail speculation floods into alts last, chasing exponential returns, often marking **distribution/bubble phase**.

4. Bear Market (BTC Regains Dominance)

- As liquidity drains and speculation dies, alts bleed harder than BTC.
- BTC dominance **rises again**, since BTC is seen as the "safe haven" within crypto.
- This marks **Stage 4 (markdown)** of the market cycle.

Rinse and repeat.
Where are we now? (24/08/2025)

0.0.2 MVRV-Z Score

MVRV-Z score: Market Value / Realized Value – Z core

In essence it is $MVRV\text{-}Z = \dfrac{\text{Market Value} - \text{Realised Value}}{\text{St.d of Market Value}}$

As the name states: It's the difference between market value and realized value. It indicates whether on average holders are in profit or loss and then normalizes by dividing by st.d (excludes anomalies)

Realized Value vs. Market Value:

- Market Value: The current spot pricen that BTC is being traded at
- Realized Value: Maintains the price that the BTC had at last on-chain movement.
  F.ex
  - You bought 1 BTC at $10k and haven't moved it since.
  - Someone else bought 1 BTC at $50k and still holds it.
  - If the spot price today is $60k, **Market Cap** would count both coins at $60k each ($120k total).
  - But **Realized Value** would record them at $10k + $50k = **$60k total**.

In essence, MVRV-Z scores give an indication of holders in profit vs. in loss.

- MV > RV → Holders are in Profit
- MV < RV → Holders are in Loss

Historical Trends for MVRV-Z Score and alignment with market cycles:

Red zone (Z > 7–9):

- Historically aligns with **market cycle tops** (2011, 2013, 2017, 2021).
- Suggests Bitcoin is significantly overvalued relative to on-chain cost basis.

Green zone (Z < 0):

- Historically aligns with **cycle bottoms** (2011, 2015, 2018, 2020).
- Suggests market participants are holding coins worth less than they paid for them → capitulation, undervaluation.

Phases in the Market Cycle (MVRV-Z behavior)

Accumulation Phase (Bear Market Bottoms)

- MVRV-Z dips into **green zone (<0)**.
- Coins are trading below cost basis.
- Long-term holders accumulate.

Expansion Phase (Early Bull)

- MVRV-Z climbs out of the green zone, moving between **1–4**.
- Still healthy, undervaluation ending.
- "Smart money" rotates in, but no extreme euphoria yet.

Distribution Phase (Late Bull / Bubble)

- MVRV-Z spikes into **red zone** (>**7–9**).
- Holders have large unrealized profits.
- Retail FOMO enters, historically near **cycle tops**.

Decline Phase (Bear Market)

- MVRV-Z falls sharply as price corrects.
- Moves back down toward neutral (∼1).

Eventually dips into **green zone** (<**0**) again → resetting the cycle

Other indicators: Puell Multiple, Logarithmic Regression Band Chart and Pi Indicator

Puell Multiple

What it measures:

- Looks at miner revenue in USD terms relative to its long-term average.
- Miner revenue = number of BTC mined daily × price of BTC.

- **Formula:** $Puell = \dfrac{\text{Daily BTC Issuance (USD)}}{\text{365-day moving average of issuance (USD)}}$

Formula Explanation: Numerator – current miner revenue: Daily BTC Issuance = BTC Mined × BTC Price Denominator – 365 Day Average of Daily BTC Issuance

- Indicates the extent to which miners are profiting over the average.

Interpretation across cycles:

- High (>4–5): Miners making outsized profits → historically near cycle tops (miners may start selling heavily).
  - Peaks around 8
- Low (<0.5): Miner revenue unusually low → stress for miners, potential bottom signals (capitulation).

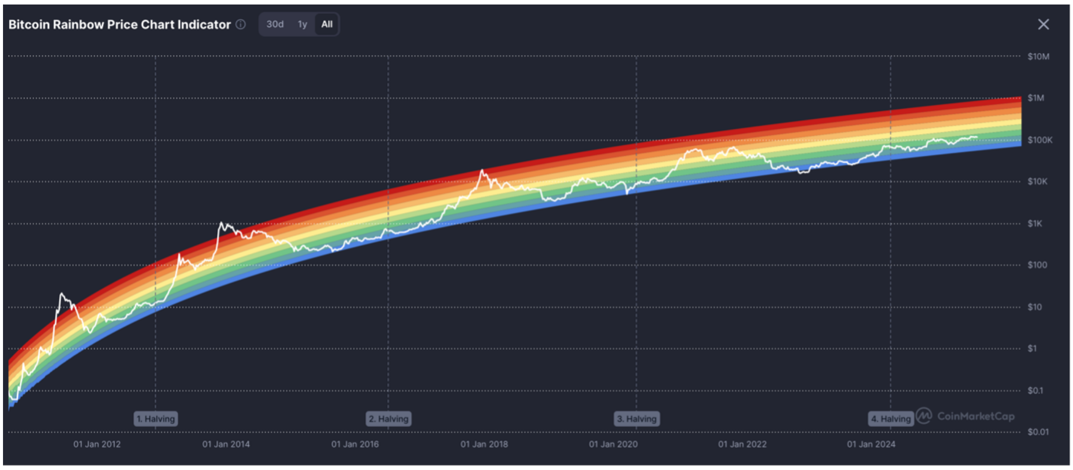
In different phases:

- Accumulation: Puell < 0.5–1, miners under stress, weak revenue.
- Expansion: Revenue improves, Puell rises toward 1–2.
- Distribution (top): Revenue surges (from both price increase and issuance), Puell spikes >4.
- Capitulation: Price crash reduces miner revenue → Puell falls back under 1.

Puell Vs. MVRV-Z Score:

- MVRV-Z – Indicates Investor Profitability
- Puell Multiple – Indicates Miner Profitability

BTC Logarithmic Regression Band Chart – Also used to determine peaks/tops:



Useful in giving general impression of how heated market is – but lots of shortcomings:

Shortcomings of Rainbow Chart:

1.Diminishing Returns Over Time

- Early cycles (2011–2013, 2015–2017) saw exponential gains because Bitcoin was small, illiquid, and more speculative.
- As Bitcoin grows in adoption and liquidity, its ability to increase 10x or 100x per cycle diminishes.
- Each successive cycle has seen:
  - Lower percentage increases (e.g., ∼100x in 2011 vs. ∼20x in 2017 vs. ∼6x in 2021).
  - Longer consolidation phases.
- The rainbow chart, by projecting similar exponential curves into the future, ignores this diminishing return effect.

2. Market Cap and Capital Inflow Problem

- To push BTC from \$10B → \$100B required ∼\$90B in inflows.
- To move BTC from \$1T → \$10T requires ∼\$9T in inflows.
- Larger market cap = exponentially more capital needed for similar % gains.
- The rainbow chart doesn't account for the reality of scaling: it assumes the same growth trajectory, which is unlikely as Bitcoin matures and global liquidity constraints come into play.

3. Regression-Based, Not Causal

- The rainbow chart is based on a logarithmic regression fit of past Bitcoin price data.
- Problem: Past price behavior may not repeat in the same way as Bitcoin matures, institutions enter, and macro cycles change.
- Moreover, different analysts will have different regression bands causing slight discrepancy

4. Upper and Lower Bands Are Arbitrary

- There is no statistical or fundamental basis for where the lines are drawn — they simply follow historical volatility.
- This means future cycles could:
  - Overshoot the top band (like in 2011 and 2013).

– Fail to reach the upper bands at all (possibly in future cycles).

- As a result, the predictive power is weak — it gives broad context but not actionable price targets.

Pi Cycle Top Indicator: Pi Multiple

Pi multiple is a BTC top indicator based off the 111 Day SMA and the 350 Day SMA x 2

- In essence: When the 111 Day SMA crosees the 350 Day SMA $\times$ 2 , it coincides with historical market tops
- Pi Cycle because : $350/111 = 3.1415$

Formula

- Fast Line (FL) = 111-day SMA (average BTC price over last 111 days)
- Slow Line (SL) = 350-day SMA $\times$ 2

Pi Cycle Indicator Trigger = FL crosses above SL.

Interpretation

- Cycle Tops: When 111-day SMA crosses above 2$\times$ 350-day SMA $\rightarrow$ historically signals overheated market and imminent top.
- Cycle Bottoms: Not designed for bottoms (it's a top indicator, unlike Puell or MVRV which also give bottom signals).
- Between Crossovers: Market phases unfold normally (accumulation, growth, euphoria, distribution), but traders watch for the next crossover as a macro warning.

LIMITATIONS:

- Based only on past prices $\rightarrow$ Can give false signals
- Maturing Markets $\rightarrow$ Volatility decreases $\rightarrow$ SMA relationships may not play out